

Advanced Java Programming

Assignment No. 3 – Unit 3: Java Server Pages (JSP)

JSP Tag Elements Quick Reference:

Tag Type	Syntax	Purpose
Expression	<code><%= expression %></code>	Outputs value directly to browser
Scriptlet	<code><% java code %></code>	Embeds Java code in JSP
Declaration	<code><%! method/variable %></code>	Declares methods or instance variables
Directive	<code><%@ directive %></code>	Page settings, includes, taglibs
Action	<code><jsp:action .../></code>	useBean, setProperty, getProperty, forward, include
Comment	<code><%-- comment --%></code>	JSP comment (not sent to client)

Q1. JSP page to display "Hello World" using expression tag `<%= %>`.

Concept:

The **Expression tag** `<%= %>` evaluates a Java expression and sends its result (as String) directly to the browser output. No semicolon is used inside expression tags.

```
<%-- File: hello.jsp --%>
<%@ page language="java" contentType="text/html; charset=UTF-8" %>
<!DOCTYPE html>
<html>
<head>
    <title>Hello World JSP</title>
    <style>
        body { font-family: Arial; text-align: center; margin-top: 50px; }
        h1 { color: #1a237e; }
    </style>
</head>
<body>
    <%-- Expression tag: outputs value directly to browser --%>
    <h1><%= "Hello, World!" %></h1>

    <%-- More expression examples --%>
    <p>Simple Math : <%= 10 + 20 %></p>
    <p>String Length: <%= "Advanced Java".length() %></p>
    <p>Uppercase      : <%= "hello".toUpperCase() %></p>
    <p>Today's Date   : <%= new java.util.Date() %></p>
</body>
</html>
```

Browser Output:

```
Hello, World!
Simple Math : 30
String Length: 13
Uppercase   : HELLO
Today's Date : Sat Mar 07 10:30:00 IST 2026
```

Note: Expression tag `<%= expr %>` is equivalent to `out.print(expr)` in a scriptlet. No semicolon needed inside `<%= %>`.

Q2. JSP page to display the current date and time using a scriptlet.

Concept:

The **Scriptlet tag** `<% %>` allows embedding Java code inside JSP. You can import classes using the **page directive** and use the **out** implicit object to print output.

```

<%-- File: datetime.jsp --%>
<%@ page language="java" contentType="text/html; charset=UTF-8"
    import="java.util.Date, java.text.SimpleDateFormat" %>
<!DOCTYPE html>
<html>
<head><title>Date and Time</title></head>
<body style="font-family:Arial; margin:30px;">
    <h2>Current Date and Time</h2>

    <%-- Scriptlet: Java code block --%>
    <%
        // Create Date object
        Date now = new Date();

        // Format date
        SimpleDateFormat dateFormat = new SimpleDateFormat("dd/MM/yyyy");
        SimpleDateFormat timeFormat = new SimpleDateFormat("HH:mm:ss");
        SimpleDateFormat dayFormat = new SimpleDateFormat("EEEE");
        SimpleDateFormat fullFormat = new SimpleDateFormat("dd MMMM yyyy, hh:mm:ss a");

        String formattedDate = dateFormat.format(now);
        String formattedTime = timeFormat.format(now);
        String dayName      = dayFormat.format(now);
        String fullDateTime = fullFormat.format(now);

    %>

    <%-- Display using expression tags --%>
    <table border="1" cellpadding="8" style="border-collapse:collapse;">
        <tr style="background:#1a237e; color:white;">
            <th>Format</th><th>Value</th>
        </tr>
        <tr><td>Date (dd/MM/yyyy)</td><td><b><%= formattedDate %></b></td></tr>
        <tr><td>Time (HH:mm:ss)</td><td><b><%= formattedTime %></b></td></tr>
        <tr><td>Day Name</td><td><b><%= dayName %></b></td></tr>
        <tr><td>Full Date & Time</td><td><b><%= fullDateTime %></b></td></tr>
        <tr><td>Milliseconds</td><td><b><%= now.getTime() %></b></td></tr>
    </table>

    <% // You can also write directly using out object
        out.println("<p>Page loaded at: " + now + "</p>");
    %>
</body>
</html>

```

Browser Output:

Format	Value
Date (dd/MM/yyyy)	07/03/2026
Time (HH:mm:ss)	10:30:45
Day Name	Saturday
Full Date & Time	07 March 2026, 10:30:45 AM
Milliseconds	1741333845000

Q3. JSP page using declarative tag to declare a method that returns a greeting message.

Concept:

The **Declaration tag** `<%! %>` declares methods and instance variables. These become part of the servlet class (not inside service() method), so they can be called from anywhere in the JSP. Unlike scriptlets, declarations are class-level.

```

<%-- File: declaration.jsp --%>
<%@ page language="java" contentType="text/html; charset=UTF-8"
    import="java.util.Calendar" %>
<!DOCTYPE html>

```

```

<html>
<head><title>Declaration Tag Demo</title></head>
<body style="font-family:Arial; margin:30px;">

    <!-- DECLARATION TAG: Declares methods and instance variables -->
    <%!
        // Instance variable (shared across requests - be careful!)
        private int visitCount = 0;

        // Method: returns greeting based on time of day
        public String getGreeting(String name) {
            Calendar cal = Calendar.getInstance();
            int hour = cal.get(Calendar.HOUR_OF_DAY);
            String msg;

            if (hour >= 5 && hour < 12)
                msg = "Good Morning";
            else if (hour >= 12 && hour < 17)
                msg = "Good Afternoon";
            else if (hour >= 17 && hour < 21)
                msg = "Good Evening";
            else
                msg = "Good Night";

            return msg + ", " + name + "!";
        }

        // Method: returns formatted string
        public String formatName(String first, String last) {
            return last.toUpperCase() + ", " + first;
        }

        // Method: calculate square
        public int square(int n) {
            return n * n;
        }

        // Method: increment and return visit count
        public int incrementVisit() {
            return ++visitCount;
        }
    %>

    <h2>Declaration Tag Demo</h2>

    <!-- Call the declared methods using Expression tags -->
    <p>Greeting      : <b><%= getGreeting("Rahul") %></b></p>
    <p>Full Name     : <b><%= formatName("Rahul", "Sharma") %></b></p>
    <p>Square of 7    : <b><%= square(7) %></b></p>
    <p>Square of 12   : <b><%= square(12) %></b></p>
    <p>Visit Count    : <b><%= incrementVisit() %></b></p>

</body>
</html>

```

Scriptlet vs Declaration Comparison:

```

<%! int x = 10; %>    <- Declaration: x is an INSTANCE VARIABLE of servlet class
<% int y = 20; %>    <- Scriptlet: y is a LOCAL VARIABLE inside service() method

```

Declaration methods can be called from Expression tags: <%= getGreeting("name") %>
 Scriptlet code runs sequentially inside service() method.

Q4. JSP page to demonstrate the use of request and response implicit objects.

Concept:

JSP provides **9 implicit objects** that are available without declaration. **request** (HttpServletRequest) gives access to form parameters, headers, attributes. **response** (HttpServletResponse) controls the HTTP response.

HTML Form (requestDemo.html):

```
<!DOCTYPE html>
<html>
<body style="font-family:Arial; max-width:400px; margin:50px auto;">
  <h2>Request Demo Form</h2>
  <form action="requestDemo.jsp" method="POST">
    Name : <input type="text" name="name" value="Rahul"><br><br>
    City : <input type="text" name="city" value="Pune"><br><br>
    Course: <input type="text" name="course" value="MCA"><br><br>
    <input type="submit" value="Submit">
  </form>
</body>
</html>
```

requestDemo.jsp:

```
<%-- File: requestDemo.jsp --%>
<%@ page language="java" contentType="text/html; charset=UTF-8"
    import="java.util.Enumeration" %>
<!DOCTYPE html>
<html>
<head><title>Request & Response Demo</title></head>
<body style="font-family:Arial; margin:20px;">

  <%-- Set response properties --%>
  <% response.setHeader("Cache-Control", "no-cache"); %>

  <h2>Request Implicit Object Demo</h2>

  <%-- Reading form parameters --%>
  <h3>Form Parameters:</h3>
  <p>Name : <b><%= request.getParameter("name") %></b></p>
  <p>City : <b><%= request.getParameter("city") %></b></p>
  <p>Course : <b><%= request.getParameter("course") %></b></p>

  <%-- Request information --%>
  <h3>Request Information:</h3>
  <p>HTTP Method : <b><%= request.getMethod() %></b></p>
  <p>Request URI : <b><%= request.getRequestURI() %></b></p>
  <p>Protocol : <b><%= request.getProtocol() %></b></p>
  <p>Server Name : <b><%= request.getServerName() %></b></p>
  <p>Server Port : <b><%= request.getServerPort() %></b></p>
  <p>Client IP : <b><%= request.getRemoteAddr() %></b></p>
  <p>Content Type : <b><%= request.getContentType() %></b></p>

  <%-- Request Headers --%>
  <h3>Request Headers:</h3>
  <%
    Enumeration<String> headers = request.getHeaderNames();
    while (headers.hasMoreElements()) {
      String headerName = headers.nextElement();
      out.println("<p>" + headerName + " : <b>" +
        request.getHeader(headerName) + "</b></p>");
    }
  %>

  <h3>Response Object Demo:</h3>
  <p>Response Content Type : <b><%= response.getContentType() %></b></p>
  <p>Buffer Size : <b><%= response.getBufferSize() %> bytes</b></p>

  <%-- Set an attribute in request and retrieve it --%>
  <% request.setAttribute("message", "Hello from JSP!"); %>
```

[illegible]

```

        if (application.getAttribute("appStartTime") == null)
            application.setAttribute("appStartTime", new Date().toString());
    %>

<h2>Application (ServletContext) Object</h2>
<p>Total Page Hits : <b><%= application.getAttribute("hitCount") %></b></p>
<p>App Start Time : <b><%= application.getAttribute("appStartTime") %></b></p>
<p>Server Info : <b><%= application.getServerInfo() %></b></p>
<p>Servlet API Ver : <b><%= application.getMajorVersion() %>.<%= application.getMinorVersion() %></b></p>
<p>Context Path : <b><%= application.getContextPath() %></b></p>
<p>Real Path(/WEB-INF): <b><%= application.getRealPath("/WEB-INF") %></b></p>

<hr>
<h3>Scope Comparison:</h3>
<table border="1" cellpadding="6" style="border-collapse:collapse;">
    <tr style="background:#1a237e;color:white;">
        <th>Scope</th><th>Object</th><th>Lifetime</th><th>Shared By</th>
    </tr>
    <tr><td>Page</td>        <td>pageContext</td>    <td>Current JSP only</td>    <td>Only current request on this p
    <tr><td>Request</td>      <td>request</td>        <td>One request</td>        <td>Pages in forward/include chain
    <tr><td>Session</td>      <td>session</td>          <td>User session</td>      <td>All pages for one user</td></tr>
    <tr><td>Application</td><td>application</td>    <td>App lifetime</td>      <td>ALL users of the application</td>
</table>

</body>
</html>

```

Q6. JSP page to demonstrate exception implicit object handling by generating an exception.

Concept:

The **exception** implicit object is available only on **error pages** (pages with `isErrorPage="true"`). When any JSP throws an exception and has `errorPage` set, JSP container forwards to error page automatically.

Main JSP (divisionPage.jsp) — generates exception:

```

<%-- File: divisionPage.jsp --%>
<%@ page language="java" contentType="text/html; charset=UTF-8"
    errorPage="errorHandler.jsp" %>
<!DOCTYPE html>
<html>
<head><title>Division Demo</title></head>
<body style="font-family:Arial; margin:30px;">
    <h2>Division Calculator</h2>

    <%
        int a = 100;
        int b = 0; // Intentionally 0 to cause ArithmeticException

        // This will throw java.lang.ArithmeticException: / by zero
        int result = a / b;
    %>

    <%-- This line will NOT be reached due to exception above --%>
    <p>Result: <%= result %></p>

</body>
</html>

```

Error Page (errorHandler.jsp) — handles exception:

```

<%-- File: errorHandler.jsp --%>
<%@ page language="java" contentType="text/html; charset=UTF-8"
    isErrorPage="true"
    import="java.io.PrintWriter, java.io.StringWriter" %>
<!DOCTYPE html>
<html>

```

```

<head>
  <title>Error Page</title>
  <style>
    body { font-family:Arial; margin:30px; }
    .error-box { background:#ffebee; border:2px solid #f44336;
                  padding:20px; border-radius:8px; }
    h2 { color:#b71c1c; }
  </style>
</head>
<body>
  <div class="error-box">
    <h2>An Error Occurred!</h2>

    <!-- exception is the implicit object available on error pages -->
    <p><b>Exception Type :</b> <%= exception.getClass().getName() %></p>
    <p><b>Error Message :</b> <%= exception.getMessage() %></p>

    <!-- Get stack trace as string -->
    <%
      StringWriter sw = new StringWriter();
      exception.printStackTrace(new PrintWriter(sw));
      String stackTrace = sw.toString();
    %>
    <h3>Stack Trace:</h3>
    <pre style="background:#fff; padding:10px; overflow:auto;">
<%= stackTrace %>
    </pre>

    <p><a href="divisionPage.jsp">Try Again</a> | <a href="index.jsp">Home</a></p>
  </div>
</body>
</html>

```

Browser Output on errorHandler.jsp:

```

An Error Occurred!
Exception Type : java.lang.ArithmeticException
Error Message  : / by zero

```

```

Stack Trace:
java.lang.ArithmeticException: / by zero
    at org.apache.jsp.divisionPage_jsp._jspService(dimensionPage_jsp.java:78)
    at org.apache.jasper.runtime.HttpJspBase.service(HttpJspBase.java:70)
    ...

```

Note: `errorPage` in calling JSP must match exactly the path of the error page. `isErrorPage='true'` enables the 'exception' implicit object.

Q7. Create a JavaBean with name and age properties and display them in a JSP using `jsp:useBean` and `jsp:getProperty`.

Concept:

A **JavaBean** is a plain Java class with: private fields, public getter/setter methods, and a no-argument constructor. JSP uses `<jsp:useBean>` to create/use a bean and `<jsp:getProperty>` to read its values.

StudentBean.java (JavaBean class):

```
// File: StudentBean.java (place in WEB-INF/classes or a package)
public class StudentBean {

    // Private fields (properties)
    private String name = "Not Set";
    private int age = 0;
    private String course = "Not Set";

    // No-argument constructor (REQUIRED for JavaBean)
    public StudentBean() { }

    // Getter and Setter for name
    public String getName() { return name; }
    public void setName(String n) { this.name = n; }

    // Getter and Setter for age
    public int getAge() { return age; }
    public void setAge(int a) { this.age = a; }

    // Getter and Setter for course
    public String getCourse() { return course; }
    public void setCourse(String c){ this.course = c; }
}
```

displayBean.jsp:

```
<%-- File: displayBean.jsp --%>
<%@ page language="java" contentType="text/html; charset=UTF-8" %>
<!DOCTYPE html>
<html>
<head><title>JavaBean Demo</title></head>
<body style="font-family:Arial; margin:30px;">

    <%-- jsp:useBean: creates or finds a bean in specified scope --%>
    <jsp:useBean id="student" class="StudentBean" scope="session" />

    <%-- Set properties using scriptlet (for demonstration) --%>
    <%
        student.setName("Rahul Sharma");
        student.setAge(21);
        student.setCourse("MCA");
    %>

    <h2>Student Information (JavaBean)</h2>

    <table border="1" cellpadding="8" style="border-collapse:collapse;">
        <tr style="background:#1a237e; color:white;">
            <th>Property</th><th>Value</th>
        </tr>
        <tr>
            <td>Name</td>
            <%-- jsp:getProperty reads the property using getter method --%>
            <td><b><jsp:getProperty name="student" property="name" /></b></td>
        </tr>
        <tr>
```



```

        <td>Age</td>
        <td><b><jsp:getProperty name="student" property="age" /></b></td>
    </tr>
    <tr>
        <td>Course</td>
        <td><b><jsp:getProperty name="student" property="course" /></b></td>
    </tr>
</table>

<!-- You can also access bean properties using EL (Expression Language) -->
<h3>Using Expression Language (EL):</h3>
<p>Name   : <b>${student.name}</b></p>
<p>Age    : <b>${student.age}</b></p>
<p>Course : <b>${student.course}</b></p>

</body>
</html>

```

Browser Output:

Student Information (JavaBean)

Property	Value
Name	Rahul Sharma
Age	21
Course	MCA

Using Expression Language (EL):

Name : Rahul Sharma

Age : 21

Course: MCA

Note: JavaBean scope can be: page (current page only), request (current request), session (user session), application (all users).

Q8. JSP page to set values in a JavaBean using jsp:setProperty and display them.

Concept:

<jsp:setProperty> sets bean properties. It can set: a specific value, from a request parameter with same name, or all parameters at once using property="*".

```

<!-- File: setPropertyDemo.jsp -->
<%@ page language="java" contentType="text/html; charset=UTF-8" %>
<!DOCTYPE html>
<html>
<head><title>setProperty Demo</title></head>
<body style="font-family:Arial; margin:30px;">

    <!-- Step 1: Create the bean -->
    <jsp:useBean id="student" class="StudentBean" scope="request" />

    <!-- Step 2: Set individual properties with hardcoded values -->
    <jsp:setProperty name="student" property="name" value="Priya Patil" />
    <jsp:setProperty name="student" property="age" value="22" />
    <jsp:setProperty name="student" property="course" value="BCA" />

    <!-- Step 3: Display using getProperty -->
    <h2>Method 1: setProperty with hardcoded value</h2>
    <p>Name : <jsp:getProperty name="student" property="name" /></p>
    <p>Age : <jsp:getProperty name="student" property="age" /></p>
    <p>Course: <jsp:getProperty name="student" property="course" /></p>

    <hr>
    <!-- Method 2: Set from request parameter (URL: ?name=Amit&age=23&course=MCA) -->
    <jsp:useBean id="student2" class="StudentBean" scope="request" />
    <!-- Set from request parameter with same name as property -->

```

```

<jsp:setProperty name="student2" property="name" param="name" />
<jsp:setProperty name="student2" property="age" param="age" />
<jsp:setProperty name="student2" property="course" param="course" />

<h2>Method 2: setProperty from Request Parameter</h2>
<p>Name : <jsp:getProperty name="student2" property="name" /></p>
<p>Age : <jsp:getProperty name="student2" property="age" /></p>
<p>Course: <jsp:getProperty name="student2" property="course" /></p>

<hr>
<!-- Method 3: Set ALL properties at once using property="*"
      (property names must match request parameter names exactly) -->
<jsp:useBean id="student3" class="StudentBean" scope="request" />
<jsp:setProperty name="student3" property="*" />

<h2>Method 3: setProperty with property="*" (all params)</h2>
<p>Name : <jsp:getProperty name="student3" property="name" /></p>
<p>Age : <jsp:getProperty name="student3" property="age" /></p>
<p>Course: <jsp:getProperty name="student3" property="course" /></p>

</body>
</html>

```

Note: `property=*` automatically maps ALL request parameters to bean properties with matching names. This is the most convenient way for form submissions.

Q9. JSP page to demonstrate JavaBean in a form submission scenario.

HTML Form (studentForm.html):

```

<!DOCTYPE html>
<html>
<body style="font-family:Arial; max-width:400px; margin:50px auto;">
  <h2>Student Registration</h2>
  <form action="processStudent.jsp" method="POST">
    Name : <input type="text" name="name" required><br><br>
    Age : <input type="number" name="age" required><br><br>
    Course : <select name="course">
      <option>BCA</option>
      <option>MCA</option>
      <option>BSc CS</option>
    </select><br><br>
    Email : <input type="email" name="email" required><br><br>
    <input type="submit" value="Register">
  </form>
</body>
</html>

```

StudentBean.java (updated with email):

```

public class StudentBean {
  private String name = "";
  private int age = 0;
  private String course = "";
  private String email = "";

  public StudentBean() {}

  // Getters
  public String getName() { return name; }
  public int getAge() { return age; }
  public String getCourse() { return course; }
  public String getEmail() { return email; }

  // Setters
  public void setName(String name) { this.name = name; }
  public void setAge(int age) { this.age = age; }

```

```

    public void setCourse(String course) { this.course = course; }
    public void setEmail(String email) { this.email = email; }
}

```

processStudent.jsp:

```

<!-- File: processStudent.jsp -->
<%@ page language="java" contentType="text/html; charset=UTF-8" %>
<!DOCTYPE html>
<html>
<head><title>Student Registration Result</title></head>
<body style="font-family:Arial; margin:30px;">

    <!-- Create bean and auto-populate from form using property="*" -->
    <jsp:useBean id="student" class="StudentBean" scope="session" />
    <jsp:setProperty name="student" property="*" />

    <!-- Basic validation using scriptlet -->
    <%
        String name = student.getName();
        int age = student.getAge();
        String course = student.getCourse();
        String email = student.getEmail();

        boolean isValid = (name != null && !name.isEmpty() &&
            age > 0 && age < 100 &&
            email != null && email.contains("@"));
    %>

    <% if (isValid) { %>
        <div style="background:#e8f5e9; border:2px solid #4caf50;
            padding:20px; border-radius:8px;">
            <h2 style="color:green">Registration Successful!</h2>
            <table border="1" cellpadding="8" style="border-collapse:collapse; width:50%;">
                <tr style="background:#1a237e; color:white;">
                    <th>Field</th><th>Value</th>
                </tr>
                <tr><td>Name</td>
                    <td><jsp:getProperty name="student" property="name" /></td></tr>
                <tr><td>Age</td>
                    <td><jsp:getProperty name="student" property="age" /></td></tr>
                <tr><td>Course</td>
                    <td><jsp:getProperty name="student" property="course"/></td></tr>
                <tr><td>Email</td>
                    <td><jsp:getProperty name="student" property="email" /></td></tr>
            </table>
            <p><a href="studentForm.html">Register Another</a></p>
        </div>
    <% } else { %>
        <div style="background:#ffebee; border:2px solid #f44336; padding:20px;">
            <h2 style="color:red">Validation Error!</h2>
            <p>Please fill all fields correctly.</p>
            <a href="studentForm.html">Go Back</a>
        </div>
    <% } %>

</body>
</html>

```

■■ JSTL (JSP Standard Tag Library) Programs ■■

JSTL requires adding **jstl.jar** and **standard.jar** to WEB-INF/lib. Use `<%@ taglib %>` directive to import tag libraries.

```
Core Tags      : <%@ taglib prefix="c"      uri="http://java.sun.com/jsp/jstl/core"      %>
Format Tags    : <%@ taglib prefix="fmt"    uri="http://java.sun.com/jsp/jstl/fmt"    %>
SQL Tags       : <%@ taglib prefix="sql"    uri="http://java.sun.com/jsp/jstl/sql"    %>
XML Tags       : <%@ taglib prefix="x"      uri="http://java.sun.com/jsp/jstl/xml"    %>
Functions      : <%@ taglib prefix="fn"    uri="http://java.sun.com/jsp/jstl/functions" %>
```

Q10. JSP page using JSTL core tag to display a message if a number is positive.

```
<%-- File: positiveNumber.jsp --%>
<%@ page language="java" contentType="text/html; charset=UTF-8" %>
<%@ taglib prefix="c" uri="http://java.sun.com/jsp/jstl/core" %>
<!DOCTYPE html>
<html>
<head><title>JSTL if Tag Demo</title></head>
<body style="font-family:Arial; margin:30px;">

    <h2>JSTL &lt;c:if&gt; Tag Demo</h2>

    <%-- Set a variable using c:set --%>
    <c:set var="number" value="25" />

    <p>Number: <b>${number}</b></p>

    <%-- c:if: single condition check (no else) --%>
    <c:if test="${number > 0}">
        <p style="color:green"><b>The number ${number} is POSITIVE.</b></p>
    </c:if>

    <c:if test="${number < 0}">
        <p style="color:red"><b>The number ${number} is NEGATIVE.</b></p>
    </c:if>

    <c:if test="${number == 0}">
        <p style="color:blue"><b>The number is ZERO.</b></p>
    </c:if>

    <%-- Another example with different numbers --%>
    <h3>Multiple Number Tests:</h3>
    <c:forEach var="n" items="15, -5, 0, 42, -100">
        <c:if test="${n > 0}">
            <p>${n} is <span style="color:green">Positive</span></p>
        </c:if>
        <c:if test="${n < 0}">
            <p>${n} is <span style="color:red">Negative</span></p>
        </c:if>
        <c:if test="${n == 0}">
            <p>${n} is <span style="color:blue">Zero</span></p>
        </c:if>
    </c:forEach>

</body>
</html>
```

Output:

Number: 25
The number 25 is POSITIVE.

Multiple Number Tests:
15 is Positive

-5 is Negative
0 is Zero
42 is Positive
-100 is Negative

Q11. JSP page using JSTL tag to display "Pass" or "Fail" based on a marks value.

Concept:

<c:choose>, **<c:when>**, **<c:otherwise>** work like if-else if-else in Java. Used for multiple conditional checks.

```
<%-- File: passFail.jsp --%>
<%@ page language="java" contentType="text/html; charset=UTF-8" %>
<%@ taglib prefix="c" uri="http://java.sun.com/jsp/jstl/core" %>
<!DOCTYPE html>
<html>
<head><title>Pass/Fail JSTL Demo</title></head>
<body style="font-family:Arial; margin:30px;">

    <h2>Student Result System using JSTL</h2>

    <%-- Set marks variable --%>
    <c:set var="marks" value="72" />

    <p>Marks Obtained: <b>${marks} / 100</b></p>

    <%-- c:choose = switch/if-else if-else --%>
    <c:choose>
        <c:when test="${marks} >= 75">
            <p style="color:darkblue; font-size:18px;">
                <b>Result: DISTINCTION</b>
            </p>
        </c:when>
        <c:when test="${marks} >= 60">
            <p style="color:green; font-size:18px;">
                <b>Result: FIRST CLASS</b>
            </p>
        </c:when>
        <c:when test="${marks} >= 50">
            <p style="color:orange; font-size:18px;">
                <b>Result: SECOND CLASS</b>
            </p>
        </c:when>
        <c:when test="${marks} >= 35">
            <p style="color:darkorange; font-size:18px;">
                <b>Result: PASS</b>
            </p>
        </c:when>
        <c:otherwise>
            <p style="color:red; font-size:18px;">
                <b>Result: FAIL</b>
            </p>
        </c:otherwise>
    </c:choose>

    <%-- Simple pass/fail display --%>
    <p>
        Simple Pass/Fail:
        <c:choose>
            <c:when test="${marks} >= 35">
                <b style="color:green">PASS</b>
            </c:when>
            <c:otherwise>
                <b style="color:red">FAIL</b>
            </c:otherwise>
        </c:choose>
    </p>
```

</p>

</body>

</html>

Output (marks=72):

Marks Obtained: 72 / 100

Result: FIRST CLASS

Simple Pass/Fail: PASS

Q12. JSP page using JSTL tag to display elements of an array.

Concept:

<c:forEach> iterates over arrays, lists, maps, and other collections. Use **var** for loop variable, **items** for collection, **varStatus** for loop index/count.

```
<%-- File: forEachDemo.jsp --%>
<%@ page language="java" contentType="text/html; charset=UTF-8"
    import="java.util.ArrayList, java.util.HashMap" %>
<%@ taglib prefix="c" uri="http://java.sun.com/jsp/jstl/core" %>
<!DOCTYPE html>
<html>
<head><title>forEach Tag Demo</title></head>
<body style="font-family:Arial; margin:30px;">

    <h2>JSTL &lt;c:forEach&gt; Demo</h2>

    <%-- Example 1: String Array --%>
    <%
        String[] fruits = {"Apple", "Banana", "Mango", "Orange", "Grapes"};
        request.setAttribute("fruits", fruits);
    %>

    <h3>1. Fruits Array:</h3>
    <ul>
    <c:forEach var="fruit" items="${fruits}">
        <li>${fruit}</li>
    </c:forEach>
    </ul>

    <%-- Example 2: With varStatus (index/count) --%>
    <h3>2. With Index (varStatus):</h3>
    <table border="1" cellpadding="6" style="border-collapse:collapse;">
        <tr style="background:#1a237e; color:white;">
            <th>Sr No</th><th>Fruit</th><th>First?</th><th>Last?</th>
        </tr>
        <c:forEach var="fruit" items="${fruits}" varStatus="status">
            <tr style="background:${status.index % 2 == 0 ? '#f5f5f5' : '#ffffff'}">
                <td>${status.count}</td>
                <td>${fruit}</td>
                <td>${status.first}</td>
                <td>${status.last}</td>
            </tr>
        </c:forEach>
    </table>

    <%-- Example 3: Integer range (begin, end, step) --%>
    <h3>3. Number Range (1 to 10, step 2):</h3>
    <c:forEach var="i" begin="1" end="10" step="2">
        <span style="margin:5px; background:#e3f2fd; padding:3px 8px;
            border-radius:4px;">${i}</span>
    </c:forEach>

    <%-- Example 4: ArrayList --%>
    <%
```

```

        ArrayList<String> courses = new ArrayList<>();
        courses.add("BCA"); courses.add("MCA");
        courses.add("BSc CS"); courses.add("BTech");
        request.setAttribute("courses", courses);
    %>
<h3>4. Course List (ArrayList):</h3>
<c:forEach var="course" items="${courses}" varStatus="s">
    <p>${s.index + 1}. ${course}</p>
</c:forEach>

</body>
</html>

```

Q13. JSP page using JSTL c:set and c:out tags to store and display a value.

```

<%-- File: setOutDemo.jsp --%>
<%@ page language="java" contentType="text/html; charset=UTF-8" %>
<%@ taglib prefix="c" uri="http://java.sun.com/jsp/jstl/core" %>
<!DOCTYPE html>
<html>
<head><title>c:set and c:out Demo</title></head>
<body style="font-family:Arial; margin:30px;">

    <h2>JSTL &lt;c:set&gt; and &lt;c:out&gt; Demo</h2>

    <%-- c:set: stores a value in a variable --%>
    <c:set var="studentName" value="Rahul Sharma" scope="page" />
    <c:set var="marks" value="85" scope="request" />
    <c:set var="course" value="Advanced Java MCA" scope="session" />

    <%-- c:out: outputs value with XSS protection (escapes HTML characters) --%>
    <p>Student Name : <b><c:out value="${studentName}" /></b></p>
    <p>Marks : <b><c:out value="${marks}" /></b></p>
    <p>Course : <b><c:out value="${course}" /></b></p>

    <%-- c:out with default value (shown when variable is null) --%>
    <p>Email : <c:out value="${email}" default="Not Provided" /></p>

    <%-- XSS protection demo --%>
    <c:set var="userInput" value="&lt;script&gt;alert('XSS')&lt;/script&gt;" />
    <p>Raw Input (dangerous): ${userInput}</p>
    <p>Safe Output (c:out) : <c:out value="${userInput}" /></p>
    <%-- c:out escapes: < > & ' " -> &lt; &gt; &amp; etc. --%>

    <%-- c:set on a JavaBean property --%>
    <jsp:useBean id="student" class="StudentBean" scope="page" />
    <c:set target="${student}" property="name" value="Priya Patil" />
    <c:set target="${student}" property="age" value="21" />
    <c:set target="${student}" property="course" value="BCA" />

    <h3>Bean values set via c:set:</h3>
    <p>Name : <c:out value="${student.name}" /></p>
    <p>Age : <c:out value="${student.age}" /></p>
    <p>Course: <c:out value="${student.course}" /></p>

    <%-- c:remove: removes a variable --%>
    <c:remove var="marks" scope="request" />
    <p>After remove - Marks: <c:out value="${marks}" default="Removed!" /></p>

</body>
</html>

```

Note: c:out is safer than \${} because it automatically escapes HTML special characters, preventing XSS (Cross-Site Scripting) attacks.

Q14. JSP page using JSTL formatting tag to display a number in currency format.

Concept:

<fmt:formatNumber> formats numbers as currency, percent, or custom patterns. **<fmt:setLocale>** sets the locale for formatting.

```
<%-- File: currencyFormat.jsp --%>
<%@ page language="java" contentType="text/html; charset=UTF-8" %>
<%@ taglib prefix="c" uri="http://java.sun.com/jsp/jstl/core" %>
<%@ taglib prefix="fmt" uri="http://java.sun.com/jsp/jstl/fmt" %>
<!DOCTYPE html>
<html>
<head><title>JSTL Formatting Demo</title></head>
<body style="font-family:Arial; margin:30px;">

    <c:set var="price" value="125000.50" />
    <c:set var="discount" value="0.15" />
    <c:set var="totalItems" value="12345678" />

    <h2>JSTL Formatting Tags Demo</h2>

    <%-- Format as Currency --%>
    <h3>Currency Formatting:</h3>

    <%-- Indian Rupee --%>
    <fmt:setLocale value="en_IN" />
    <p>Indian Rupee : <b>
        <fmt:formatNumber value="${price}" type="currency" />
    </b></p>

    <%-- US Dollar --%>
    <fmt:setLocale value="en_US" />
    <p>US Dollar : <b>
        <fmt:formatNumber value="${price}" type="currency" />
    </b></p>

    <%-- Euro --%>
    <fmt:setLocale value="de_DE" />
    <p>Euro : <b>
        <fmt:formatNumber value="${price}" type="currency" />
    </b></p>

    <%-- Percentage --%>
    <h3>Percentage Formatting:</h3>
    <fmt:setLocale value="en_US" />
    <p>Discount Rate: <b>
        <fmt:formatNumber value="${discount}" type="percent" />
    </b></p>

    <%-- Number formatting --%>
    <h3>Number Formatting:</h3>
    <p>Plain Number : <b><fmt:formatNumber value="${totalItems}" type="number" /></b></p>
    <p>With 2 decimals : <b><fmt:formatNumber value="${price}" minFractionDigits="2" maxFractionDigits="2" /></b></p>
    <p>Custom pattern : <b><fmt:formatNumber value="${price}" pattern="###,##,##0.00" /></b></p>

    <%-- Parse a number string back to number --%>
    <h3>Parsing:</h3>
    <fmt:parseNumber var="parsed" value="1,25,000.50" pattern="###,##,##0.00" />
    <p>Parsed Value: <b>${parsed}</b></p>

</body>
</html>
```

Browser Output:

Currency Formatting:

Indian Rupee : Rs.1,25,000.50
US Dollar : \$125,000.50
Euro : 125.000,50 EUR

Percentage Formatting:
Discount Rate: 15%

Number Formatting:
Plain Number : 12,345,678
With 2 decimals : 125,000.50
Custom pattern : 1,25,000.50

Q15. JSP page using JSTL tag to display the current date in dd/MM/yyyy format.

```
<%-- File: dateFormat.jsp --%>
<%@ page language="java" contentType="text/html; charset=UTF-8"
    import="java.util.Date" %>
<%@ taglib prefix="c" uri="http://java.sun.com/jsp/jstl/core" %>
<%@ taglib prefix="fmt" uri="http://java.sun.com/jsp/jstl/fmt" %>
<!DOCTYPE html>
<html>
<head><title>Date Formatting Demo</title></head>
<body style="font-family:Arial; margin:30px;">

    <%-- Store current date --%>
    <c:set var="today" value="%= new java.util.Date() %=" />

    <h2>JSTL Date Formatting Demo</h2>

    <%-- Short date format --%>
    <p>Short : <b><fmt:formatDate value="${today}" type="date" dateStyle="short" /></b></p>

    <%-- Medium date format --%>
    <p>Medium : <b><fmt:formatDate value="${today}" type="date" dateStyle="medium" /></b></p>

    <%-- Long date format --%>
    <p>Long : <b><fmt:formatDate value="${today}" type="date" dateStyle="long" /></b></p>

    <%-- Custom dd/MM/yyyy format --%>
    <p>dd/MM/yyyy : <b>
        <fmt:formatDate value="${today}" pattern="dd/MM/yyyy" />
    </b></p>

    <%-- Date and Time together --%>
    <p>Date + Time: <b>
        <fmt:formatDate value="${today}" type="both" dateStyle="medium" timeStyle="medium" />
    </b></p>

    <%-- Custom patterns --%>
    <p>Pattern 1 : <b><fmt:formatDate value="${today}" pattern="EEEE, dd MMMM yyyy" /></b></p>
    <p>Pattern 2 : <b><fmt:formatDate value="${today}" pattern="dd-MMM-yyyy HH:mm:ss" /></b></p>
    <p>Pattern 3 : <b><fmt:formatDate value="${today}" pattern="yyyy/MM/dd" /></b></p>

    <%-- Time only --%>
    <p>Time only : <b><fmt:formatDate value="${today}" type="time" timeStyle="medium" /></b></p>

</body>
</html>
```

Browser Output:

Short : 07/03/26
Medium : 07-Mar-2026
Long : 07 March 2026
dd/MM/yyyy : 07/03/2026
Date + Time: 07-Mar-2026 10:30:45 AM

Pattern 1 : Saturday, 07 March 2026
Pattern 2 : 07-Mar-2026 10:30:45
Pattern 3 : 2026/03/07
Time only : 10:30:45 AM

Q16. JSP page using JSTL XML tags to parse and display data from a small XML file.

Concept:

JSTL XML tags allow parsing XML and extracting data using **XPath** expressions. Use **<x:parse>** to parse XML and **<x:forEach>/<x:out>** to display data.

XML File (students.xml — placed in WEB-INF):

```
<?xml version="1.0" encoding="UTF-8"?>
<students>
  <student id="1">
    <name>Rahul Sharma</name>
    <age>21</age>
    <course>MCA</course>
    <marks>85</marks>
  </student>
  <student id="2">
    <name>Priya Patil</name>
    <age>20</age>
    <course>BCA</course>
    <marks>78</marks>
  </student>
  <student id="3">
    <name>Amit Joshi</name>
    <age>22</age>
    <course>BSc CS</course>
    <marks>90</marks>
  </student>
</students>
```

xmlDemo.jsp:

```
<%-- File: xmlDemo.jsp --%>
<%@ page language="java" contentType="text/html; charset=UTF-8"
    import="java.io.*" %>
<%@ taglib prefix="c" uri="http://java.sun.com/jsp/jstl/core" %>
<%@ taglib prefix="x" uri="http://java.sun.com/jsp/jstl/xml" %>
<!DOCTYPE html>
<html>
<head><title>JSTL XML Demo</title></head>
<body style="font-family:Arial; margin:30px;">

    <%-- Read XML file as string --%>
    <%
        String xmlPath = application.getRealPath("/WEB-INF/students.xml");
        BufferedReader br = new BufferedReader(new FileReader(xmlPath));
        StringBuilder sb = new StringBuilder();
        String line;
        while ((line = br.readLine()) != null) sb.append(line);
        br.close();
        String xmlData = sb.toString();
        pageContext.setAttribute("xmlData", xmlData);
    %>

    <%-- x:parse: parse the XML string --%>
    <x:parse xml="${xmlData}" var="xmlDoc" />

    <h2>JSTL XML Tag Demo</h2>

    <%-- x:out: output a single XPath value --%>
    <p>First Student: <b>
```

```
<x:out select="$xmlDoc/students/student[1]/name" />
</b></p>

<!-- x:forEach: iterate over XML nodes -->
<h3>All Students:</h3>
<table border="1" cellpadding="8" style="border-collapse:collapse;">
  <tr style="background:#1a237e; color:white;">
    <th>ID</th><th>Name</th><th>Age</th><th>Course</th><th>Marks</th>
  </tr>
  <x:forEach select="$xmlDoc/students/student" var="student">
    <tr>
      <td><x:out select="$student/@id" /></td>
      <td><x:out select="$student/name" /></td>
      <td><x:out select="$student/age" /></td>
      <td><x:out select="$student/course" /></td>
      <td><x:out select="$student/marks" /></td>
    </tr>
  </x:forEach>
</table>

<!-- x:if: conditional on XML -->
<h3>High Scorers (marks > 80):</h3>
<x:forEach select="$xmlDoc/students/student">
  <x:if select="./marks > 80">
    <p><x:out select="./name" /> - <x:out select="./marks" /></p>
  </x:if>
</x:forEach>

</body>
</html>
```

■■ JSTL SQL Tags ■■

Q17. JSP page using JSTL SQL tag to fetch all records from a database table and display them in an HTML table.

Concept:

JSTL SQL tags allow database operations directly from JSP. **<sql:setDataSource>** sets DB connection.

<sql:query> runs SELECT and returns results in a variable.

web.xml — DataSource configuration (recommended approach):

```
<!-- In web.xml: configure data source as context parameter -->
<context-param>
    <param-name>DB_URL</param-name>
    <param-value>jdbc:mysql://localhost:3306/college_db</param-value>
</context-param>
```

fetchStudents.jsp:

```
<%-- File: fetchStudents.jsp --%>
<%@ page language="java" contentType="text/html; charset=UTF-8" %>
<%@ taglib prefix="c" uri="http://java.sun.com/jsp/jstl/core" %>
<%@ taglib prefix="sql" uri="http://java.sun.com/jsp/jstl/sql" %>
<%@ taglib prefix="fmt" uri="http://java.sun.com/jsp/jstl/fmt" %>
<!DOCTYPE html>
<html>
<head>
    <title>JSTL SQL - Fetch Records</title>
    <style>
        body { font-family:Arial; margin:20px; }
        table { border-collapse:collapse; width:100%; }
        th { background:#1a237e; color:white; padding:10px; text-align:left; }
        td { padding:8px; border-bottom:1px solid #ddd; }
        tr:nth-child(even) { background:#f2f2f2; }
        tr:hover { background:#e3f2fd; }
    </style>
</head>
<body>

    <%-- Set up database connection --%>
    <sql:setDataSource
        var="db"
        driver="com.mysql.cj.jdbc.Driver"
        url="jdbc:mysql://localhost:3306/college_db"
        user="root"
        password="password" />

    <%-- Execute SELECT query --%>
    <sql:query var="result" dataSource="${db}">
        SELECT id, name, age, course FROM students ORDER BY id
    </sql:query>

    <h2>All Students</h2>
    <p>Total Records: <b>${result.rowCount}</b></p>

    <table>
        <%-- Display column headers dynamically --%>
        <tr>
            <c:forEach var="col" items="${result.columnNames}">
                <th>${col}</th>
            </c:forEach>
        </tr>

        <%-- Display rows --%>
        <c:forEach var="row" items="${result.rows}">
            <tr>
```

```

        <td>${row.id}</td>
        <td>${row.name}</td>
        <td>${row.age}</td>
        <td>${row.course}</td>
    </tr>
</c:forEach>
</table>

<%-- Display row count --%>
<p><i>Showing ${result.rowCount} student(s)</i></p>

</body>
</html>

```

Browser Output:

All Students Total Records: 3

ID	NAME	AGE	COURSE
1	Rahul Sharma	20	BCA
2	Priya Patil	21	MCA
3	Amit Joshi	22	BSc CS

Showing 3 student(s)

Note: JSTL SQL tags are convenient for simple apps, but for production, use Servlets with JDBC and connection pooling (Apache DBCP/HikariCP) for better performance and security.

Q18. JSP page using JSTL SQL tag to insert a record into a database table.

HTML Form (insertForm.html):

```

<!DOCTYPE html>
<html>
<body style="font-family:Arial; max-width:400px; margin:50px auto;">
    <h2>Add Student (JSTL SQL)</h2>
    <form action="insertStudent.jsp" method="POST">
        Name : <input type="text" name="name" required><br><br>
        Age : <input type="number" name="age" required><br><br>
        Course: <input type="text" name="course" required><br><br>
        <input type="submit" value="Insert Record">
    </form>
</body>
</html>

```

insertStudent.jsp:

```

<%-- File: insertStudent.jsp --%>
<%@ page language="java" contentType="text/html; charset=UTF-8" %>
<%@ taglib prefix="c" uri="http://java.sun.com/jsp/jstl/core" %>
<%@ taglib prefix="sql" uri="http://java.sun.com/jsp/jstl/sql" %>
<!DOCTYPE html>
<html>
<head><title>Insert Student</title></head>
<body style="font-family:Arial; margin:30px;">

    <%-- Set up database connection --%>
    <sql:setDataSource
        var="db"
        driver="com.mysql.cj.jdbc.Driver"
        url="jdbc:mysql://localhost:3306/college_db"
        user="root"
        password="password" />

    <%-- Get form parameters --%>
    <c:set var="name" value="${param.name}" />
    <c:set var="age" value="${param.age}" />

```

```

<c:set var="course" value="${param.course}" />

<!-- Execute INSERT using sql:update -->
<sql:update var="result" dataSource="${db}">
    INSERT INTO students(name, age, course) VALUES (?, ?, ?)
    <sql:param value="${name}" />
    <sql:param value="${age}" />
    <sql:param value="${course}" />
</sql:update>

<!-- Display result -->
<c:choose>
    <c:when test="${result > 0}">
        <div style="background:#e8f5e9; border:2px solid #4caf50;
            padding:20px; border-radius:8px;">
            <h2 style="color:green">Record Inserted Successfully!</h2>
            <p>Name : <b>${name}</b></p>
            <p>Age : <b>${age}</b></p>
            <p>Course: <b>${course}</b></p>
            <p>Rows affected: <b>${result}</b></p>
        </div>
    </c:when>
    <c:otherwise>
        <h2 style="color:red">Insert Failed!</h2>
    </c:otherwise>
</c:choose>

<br>
<a href="insertForm.html">Insert Another</a> |
<a href="fetchStudents.jsp">View All Records</a>

</body>
</html>

```

Note: sql:update is used for INSERT, UPDATE, DELETE. sql:query is used for SELECT. sql:param prevents SQL injection by parameterizing values.

Q19. JSP page using JSTL SQL tag to safely pass a parameter to a query.

Concept:

<sql:param> safely binds parameters to SQL queries — similar to PreparedStatement in JDBC. It prevents SQL injection by treating values as data, not SQL code. Use **<sql:dateParam>** for date parameters.

```

<!-- File: searchStudent.jsp -->
<%@ page language="java" contentType="text/html; charset=UTF-8" %>
<%@ taglib prefix="c" uri="http://java.sun.com/jsp/jstl/core" %>
<%@ taglib prefix="sql" uri="http://java.sun.com/jsp/jstl/sql" %>
<!DOCTYPE html>
<html>
<head><title>Parameterized SQL Query</title></head>
<body style="font-family:Arial; margin:30px;">

    <!-- Search Form -->
    <h2>Search Student by Course</h2>
    <form action="searchStudent.jsp" method="GET">
        Course: <input type="text" name="course" value="${param.course}">
        Min Age: <input type="number" name="minAge" value="${param.minAge}">
        <input type="submit" value="Search">
    </form>
    <hr>

    <c:if test="${not empty param.course}">

        <!-- Set up database -->
        <sql:setDataSource

```

```

        var="db"
        driver="com.mysql.cj.jdbc.Driver"
        url="jdbc:mysql://localhost:3306/college_db"
        user="root"
        password="password" />

<!-- SAFE parameterized query using sql:param -->
<sql:query var="result" dataSource="${db}">
    SELECT id, name, age, course
    FROM students
    WHERE course = ?
        AND age >= ?
    ORDER BY name
    <!-- sql:param: safely binds value to ? placeholder -->
    <sql:param value="${param.course}" />
    <sql:param value="{empty param.minAge ? 0 : param.minAge}" />
</sql:query>

<h3>Search Results for: "<b>${param.course}</b>"</h3>

<c:choose>
    <c:when test="${result.rowCount > 0}">
        <table border="1" cellpadding="8" style="border-collapse:collapse;">
            <tr style="background:#1a237e; color:white;">
                <th>ID</th><th>Name</th><th>Age</th><th>Course</th>
            </tr>
            <c:forEach var="row" items="${result.rows}">
                <tr>
                    <td>${row.id}</td>
                    <td>${row.name}</td>
                    <td>${row.age}</td>
                    <td>${row.course}</td>
                </tr>
            </c:forEach>
        </table>
        <p>Found <b>${result.rowCount}</b> student(s).</p>
    </c:when>
    <c:otherwise>
        <p style="color:red">No students found for course: ${param.course}</p>
    </c:otherwise>
</c:choose>

<!-- Demonstrate sql:dateParam -->
<!-- (Example: if table had a dateJoined column)
<sql:query var="recentStudents" dataSource="${db}">
    SELECT * FROM students WHERE dateJoined >= ?
    <sql:dateParam value="{%= new java.util.Date() %}" type="date" />
</sql:query>
--%>

</c:if>

<!-- Show why parameterized queries prevent SQL injection -->
<hr>
<h3>Why sql:param is Safe:</h3>
<table border="1" cellpadding="6" style="border-collapse:collapse; width:100%;">
    <tr style="background:#b71c1c; color:white;">
        <th>Approach</th><th>Query</th><th>Risk</th>
    </tr>
    <tr style="background:#ffebee;">
        <td>UNSAFE (string concat)</td>
        <td style="font-family:Courier; font-size:9pt;">
            WHERE course = '${param.course}'<br>
            (if input is: ' OR '1'='1 -> gets ALL records!)
        </td>

```

```

        <td>SQL Injection possible</td>
    </tr>
    <tr style="background:#e8f5e9;">
        <td>SAFE (sql:param)</td>
        <td style="font-family:Courier; font-size:9pt;">
            WHERE course = ?<br>
            &lt;sql:param value="{param.course}" /&gt;<br>
            (input is always treated as literal data)
        </td>
        <td>SQL Injection prevented</td>
    </tr>
</table>

</body>
</html>

```

Browser Output (searching for MCA):

```

Search Results for: "MCA"
ID | Name          | Age | Course
---|-----|-----|-----
2  | Priya Patil   | 21  | MCA
5  | Ravi More     | 22  | MCA

Found 2 student(s).

```